Project Report Title: Design Proposal for an Academic Research Online Intelligent Agent System

Module: IA_PCOM7E – Intelligent Agents

Assignment: Unit 6 – Development Team Project: Project Report

Team: Group A

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Person 1:

1.1 Introduction

This project develops an intelligent agent for the Academic Research Online domain. Intelligent agents are autonomous systems that perform tasks and make decisions based on defined goals (Wali, Mahamad and Sulaiman, 2023). The agent searches academic websites, retrieves relevant research information, and processes the data efficiently. This domain was chosen because researchers and students frequently need to access large volumes of academic information. Manual searching can be slow and inefficient, while automated agents can quickly identify and organise relevant research materials (Madanchian and Taherdoost, 2025).

1.2 Problem Statement

Researchers often struggle to find and organise relevant academic information online, making manual searches time-consuming (Matysik and Jaworski, 2025).

1.3 Success Criteria Mapping

The system will retrieve academic data, process it through filtering and organisation, and store results in structured formats such as CSV or JSON.

Person 2:

2.1 Functional Requirements

The system allows users to enter keywords related to academic research topics. The intelligent agent will automatically search academic platforms such as Google Scholar and retrieve relevant metadata including article titles, authors, and publication years. The

agent extracts this information and organises it into a structured dataset. The system must also remove duplicate entries and sort the collected data to support easier analysis. Finally, the processed results are stored locally in structured formats such as CSV or JSON, enabling users to access and analyse the information offline (Mitchell, 2018).

2.2 Non-functional Requirements

The system must operate reliably, accurately, and efficiently when extracting and processing data. It should also be scalable so that additional academic websites or larger datasets can be supported in the future. Furthermore, the system architecture should be modular, allowing new components or features to be integrated without major changes to the core system (Russell and Norvig, 2021).

2.3 Libraries and Tools Justification

Python is selected because of its strong capabilities in automation and data processing. BeautifulSoup and Requests will be used for web scraping, while Selenium supports interaction with dynamic webpages. Processed data will be stored using CSV or SQLite, ensuring efficient and structured data management within the agent system (Mitchell, 2018; Russell and Norvig, 2021).

Person 3:

3. Agent Model & Design Decisions

3.1 Agent Model Explanation

The system adopts Belief-Desire-Intention (BDI) model as its core agent architecture. First introduced by Bratman (1987), the model envisions autonomous systems making

use of three concepts for its functioning. The first concept is Beliefs, which represent what an agent considers to be true, with an associated degree of certainty. The second is Desires, which mean the high-level aims that an agent operates to achieve. Finally, Intentions are the concrete action plans that an agent commits to realise the desires (Perez, 2019).

Our core agent's beliefs comprise what constitutes as an academic source or an academic search engine. Beliefs also comprise the different types and components of academic sources, and how keywords can be extracted from each type. Meanwhile, the agent's desires include looking up, finding, filtering and retrieving academic sources that are the most relevant to the user's query. Finally, its intentions are to the exact plan it performs to answer the user's query and returning the results in a structure format.

3.2 Design Decisions Justification

The BDI model's separation of its three components allows the agent to reason about its tasks, adapt when expected results are unavailable, and pursue its objectives autonomously without continuous user intervention (Georgeff et al., 1999). This suits our academic online research system, as it requires a goal-oriented multi-step workflow comprising searching, extracting, processing and storing results, along with fault tolerance in the absence of relevant ones.

3.3 Weaknesses

Despite the previous strength points, some weakness points can be expected in a BDI-based agentic system. Those would mainly hover around what the agent constitutes as beliefs. A thorough build up of the agent's database of academic sources criteria and

search engines, for example, is essential for the agent's success. It follows that there might be a risk of having such a database not as comprehensive or updated as needed. Changes in academic sources search engines for example would be an update that needs to be reflected in the agent's beliefs.

Person 4:

4.1 Methodology Explanation

The Agile methodology is adopted for developing the Academic Research Online Agent. Agile is suitable for this project because it allows the system to be developed incrementally rather than all at once (Omonije, 2024). The development process is divided into smaller phases focusing on tasks such as keyword input, searching academic websites, data extraction, and storing results. This approach allows the team to test each component during development and identify issues early. It also supports continuous improvement throughout the project. In addition, modular programming is used so that different system components operate independently and can be updated or expanded in the future (Anifa et al., 2024).

4.2 Anticipated Challenges

Several challenges may arise during development. Changes in website structure can affect web scraping because updates to HTML elements may cause the extraction code to fail. Ensuring data accuracy is another challenge since some results may be duplicated or incomplete. Ethical considerations must also be addressed by respecting website policies and avoiding excessive requests. Furthermore, some websites implement rate limiting, which restricts the number of requests within a short period.

4.3 Addressing Challenges with Paradigms/Theories

These challenges can be addressed using an agent-based approach supported by the BDI model. The belief component allows the agent to update information about data sources, while intentions guide its behaviour when handling search or extraction issues (Saadi, Maamri and Sahnoun, 2020).

Person 5:

5.Graphical Designs & System Explanation

5.1 Diagram 1: Class Diagram

- The class diagram shows the main components of the Academic Research Online Agent system. The Agent acts as the central controller that coordinates all system modules.
- The Search Module performs keyword-based searches on academic websites.
- The Data Extractor retrieves metadata such as titles, authors, and publication years. The Processing Module filters and organises the collected data.
- The Storage System stores the processed results in structured formats like CSV or JSON for later use.

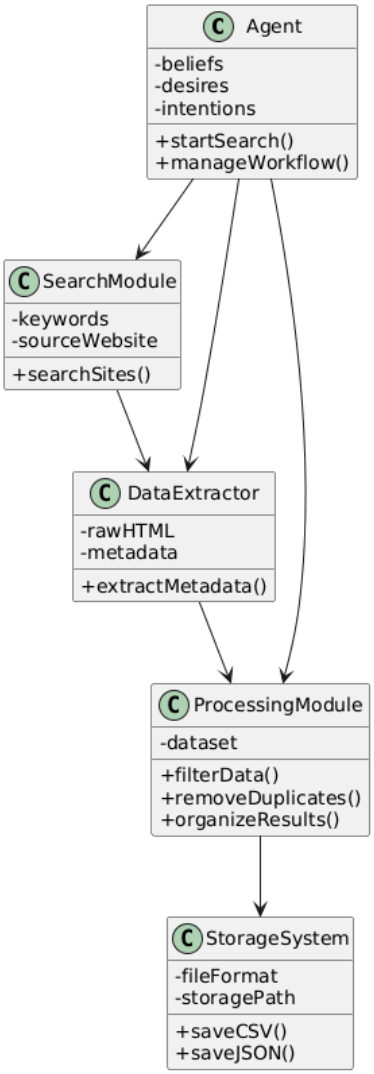

Figure 1 Class Diagram

5.2 Diagram 2: Sequence Diagram

- The sequence diagram demonstrates how the intelligent agent interacts with system modules during the research process.
- First, the User provides keywords to the Agent.
- The Agent sends the request to the Search Module, which queries academic websites.
- Retrieved data is passed to the Data Extractor, which collects metadata.
- The Processing Module then filters and organises the information.
- Finally, the Storage System saves the structured results locally.
- This sequence reflects the automated workflow described in the system design.

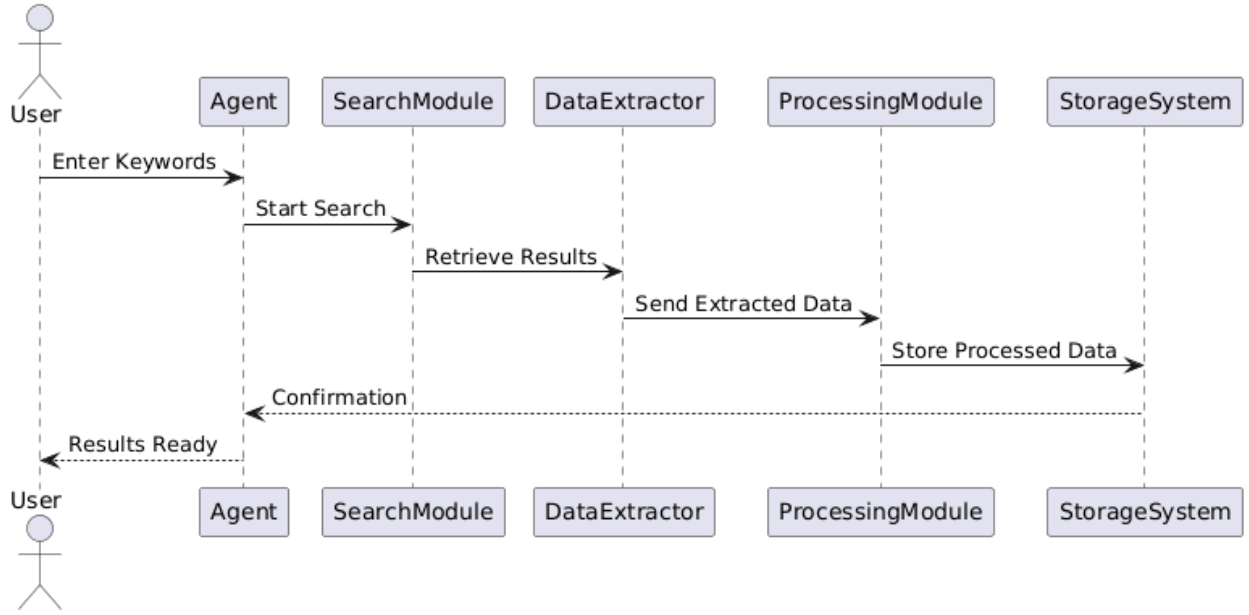

Figure 2 Sequence Diagram

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